

CompSci Graph Theory Notes

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Computer Science and Graph Theory

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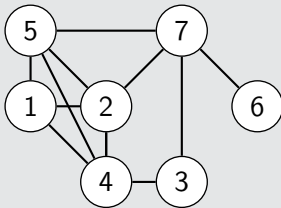
Computer Science and Graph
Theory

Computer Science and Graph Theory

- Graphs can be represented using **adjacency matrices**
- Adjacency matrices are rectangular arrays that use 1 to indicate edges in a graph

Example (Finding an adjacency matrix)

Find the adjacency matrix for the following graph:



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Example (Finding an adjacency matrix)

Find the adjacency matrix for the following graph:



- Matrices can be handled in Python using the NumPy arrays:

```
import numpy as np

A = np.array([
    [0, 1, 0, 1, 1, 0, 0],
    [1, 0, 0, 1, 1, 0, 1],
    [0, 0, 0, 1, 0, 0, 1],
    [1, 1, 1, 0, 1, 0, 0],
    [1, 1, 0, 1, 0, 0, 1],
    [0, 0, 0, 0, 0, 0, 1],
    [0, 1, 1, 0, 1, 1, 0]
])
```

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- Computer Science and Graph Theory
 - Matrices and Python

• Matrices can be handled in Python using the NumPy arrays:

```
import numpy as np

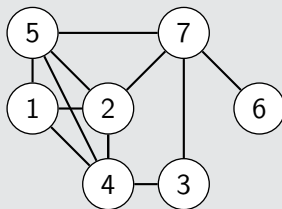
A = np.array([
    [0, 1, 0, 1, 1, 0, 0],
    [1, 0, 0, 1, 1, 0, 1],
    [0, 0, 0, 1, 0, 0, 1],
    [1, 1, 1, 0, 1, 0, 0],
    [1, 1, 0, 1, 0, 0, 1],
    [0, 0, 0, 0, 0, 0, 1],
    [0, 1, 1, 0, 1, 1, 0]
])
```

- NumPy provides many operations to use with arrays
 1. Column/row sums with `sum` (*not* `np.sum`). This is useful for finding degrees of vertices.
 2. Matrix multiplication with `@` (such as `A@A`). This is useful for counting paths between vertices.

Example (Adjacency matrices, paths and degrees)

Use the adjacency matrix to answer the following questions:

1. What is the degree of each vertex?
2. How many paths of length two are there between vertex 4 and vertex 7?
3. How many cycles of length four start at vertex 2?



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- We can draw graphs using NetworkX and matplotlib.

```
import matplotlib.pyplot as plt
import networkx as nx

G = nx.from_numpy_array(A)
nx.display(G)
plt.show()
```

- We can draw graphs using NetworkX and matplotlib.

```
import matplotlib.pyplot as plt
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```

This code should be continued from previous code!

Example (Utility graph)

Plot the utility graph using Python.

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└ Drawing Graphs Again

Example (Utility graph)

Plot the utility graph using Python.

Using the algorithm in Python

Using the provided algorithm `gale_shapley_algorithm.py`, solve the stable matching problem for the following set of preferences:

Applicant	Employer rankings	Employer	Applicant rankings
1	B, A, C	A	3, 1, 2
2	B, C, A	B	2, 3, 1
3	A, B, C	C	1, 3, 2

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